

Week-3: MongoDB BookFlow Practical

Step 1 — Create Database

```
> use BookFlow
< already on db BookFlow
> db.book_metadata.countDocuments()
< 4
```

Step 2 — Insert Book Documents

```
> db.book_metadata.find().pretty()
< {
  _id: ObjectId('6a85b3095182dc0f9d9a1bd8'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
```

>_MONGOSH

```
    member_id: 'M1001',
    rating: 5,
    comment: 'Excellent deep dive into digital systems design.',
    date: 2023-01-15T00:00:00.000Z
  },
  {
    member_id: 'M1042',
    rating: 4,
    comment: 'Good Digital edition.',
    date: 2023-02-20T00:00:00.000Z
  }
]
}
{
  _id: ObjectId('6a85b3095182dc0f9d9a1bd9'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ]
}
```

>_MONGOSH

```
],
specs: {
  pages: 210,
  fileSize_MB: 2.1,
  drm: false
},
avgRating: 4.2,
reviews: [
  {
    member_id: 'M2001',
    rating: 4,
    comment: 'Solid Digital read.',
    date: 2022-05-10T00:00:00.000Z
  },
  {
    member_id: 'M2002',
    rating: 5,
    comment: 'Best programming book.',
    date: 2022-06-01T00:00:00.000Z
  }
]
}
```

```

{
  _id: ObjectId('6a85b3095182dc0f9d9a1bda'),
  title: 'Whispers of Yesterday',
  author: 'S. Kapoor',
  genre: 'Fiction',
  publishedYear: 2019,
  formats: [
    'Hardcover',
    'Paperback'
  ],
  specs: {
    pages: 410,
    binding: 'Sewn'
  },
  avgRating: 3.8,
  reviews: [
    {
      member_id: 'M3001',
      rating: 3,
      comment: 'Slow start but strong ending.',
      date: 2020-01-15T00:00:00.000Z
    },
    {
      member_id: 'M3002',
      rating: 4,
      comment: 'Beautifully written prose.',
      date: 2020-02-11T00:00:00.000Z
    }
  ]
}

```

```

{
  _id: ObjectId('6a85b3095182dc0f9d9a1bdb'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [
    'Digital',
    'Audiobook'
  ],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: 'P. Iyer',
    runtime_min: 540
  },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'Digital format made annotations easy.',
      date: 2023-08-01T00:00:00.000Z
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: 2023-09-14T00:00:00.000Z
    }
  ]
}

```

Task 2 — Advanced Querying

Step 4 — Find Books Rated Above 4

```
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
)
< {
  title: 'Quantum Horizons',
  avgRating: 4.9
}
{
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
```

Step 5 — Find Digital/Audiobook Books

```
> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
```

Step 6 — Find Books with Rating > 4 and Digital Format

```
>_MONGODB
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
})
< {
  _id: ObjectId('6a85b3095182dc0f9d9a1bd8'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ],
}
```

```
{
  member_id: 'M1042',
  rating: 4,
  comment: 'Good Digital edition.',
  date: 2023-02-20T00:00:00.000Z
}
]
}
{
  _id: ObjectId('6a85b3095182dc0f9d9a1bd9'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ],
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read.',
      date: 2022-05-10T00:00:00.000Z
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book.',
      date: 2022-06-01T00:00:00.000Z
    }
  ]
}
```

```
{
  _id: ObjectId('6a85b3095182dc0f9d9a1bdb'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [
    'Digital',
    'Audiobook'
  ],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: 'P. Iyer',
    runtime_min: 540
  },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'Digital format made annotations easy.',
      date: 2023-08-01T00:00:00.000Z
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: 2023-09-14T00:00:00.000Z
    }
  ]
}
```

Step 7 — Regex Search for Digital Format

```
> db.book_metadata.find(
  { formats: { $regex: "Digital", $options: "i" } },
  { title: 1, formats: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
```

Step 8 — Regex Search in Review Comments

```
>_MONGOSH
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } },
  { title: 1, "reviews.$": 1 }
)
< {
  _id: ObjectId('6a85b3895182dc0f9d9a1bd8'),
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a85b3895182dc0f9d9a1bd9'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read.',
      date: 2022-05-10T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a85b3895182dc0f9d9a1bdb'),
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'Digital format made annotations easy.',
      date: 2023-08-01T00:00:00.000Z
    }
  ]
}
}
```

Task 3 — Aggregation Pipeline

Step 9 — Generate Top Rated Report

```
> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },

  { $unwind: "$reviews" },

  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },

  { $sort: { avgReviewRating: -1 } },

  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: { $round: ["$avgReviewRating", 2] },
      totalReviews: 1
    }
  }
])
```

```

< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}

```

Task 4 — Indexing

Step 10 — Create Text Index

```

> db.book_metadata.createIndex({
  "reviews.comment": "text"
})
< reviews.comment_text

```

Step 11 — Search Using Text Index

```

> .MONGOSH
> db.book_metadata.createIndex({
  "reviews.comment": "text"
})
< reviews.comment_text
> db.book_metadata.find({
  $text: { $search: "Digital" }
})
< [
  {
    _id: ObjectId('6a85b3095182dc0f9d9a1bd8'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [
      'Hardcover',
      'Digital',
      'Audiobook'
    ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: 2023-01-15T00:00:00.000Z
      }
    ]
  },

```



```

>_MONGOSH
{
  {
    member_id: 'M1042',
    rating: 4,
    comment: 'Good Digital edition.',
    date: 2023-02-20T00:00:00.000Z
  }
}
{
  _id: ObjectId('6a85b3095182dc0f9d9a1bd9'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ],
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read.',
      date: 2022-05-10T00:00:00.000Z
    },
  ],
}

```

```

>_MONGOSH
{
  {
    member_id: 'M2002',
    rating: 5,
    comment: 'Best programming book.',
    date: 2022-06-01T00:00:00.000Z
  }
}
{
  _id: ObjectId('6a85b3095182dc0f9d9a1bdb'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [
    'Digital',
    'Audiobook'
  ],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: 'P. Iyer',
    runtime_min: 540
  },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'Digital format made annotations easy.',
      date: 2023-08-01T00:00:00.000Z
    },
  ],
}

```

```

{
  {
    member_id: 'M4002',
    rating: 5,
    comment: 'Mind-bending and well narrated.',
    date: 2023-09-14T00:00:00.000Z
  }
}
}

```

Step 12 — Check Text Index Using explain()

```
> db.book_metadata.find({
  $text: { $search: "Digital" }
}).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'BookFlow.book_metadata',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [
          'digit'
        ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
      },
      textIndexVersion: 3,
      inputStage: {
        stage: 'FETCH',
```

```
>_MONGOOSH
  nss: 'BookFlow.book_metadata',
  inputStage: {
    stage: 'IXSCAN',
    nss: 'BookFlow.book_metadata',
    keyPattern: {
      _fts: 'text',
      _ftsx: 1
    },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {}
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 4,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlow.book_metadata',
    indexPrefix: {},
```

```

serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}

```

Step 13 — Create Multikey Index

```

> db.book_metadata.createIndex({ formats: 1 })
< formats_1

```

Step 14 — Create Rating Index

```

> db.book_metadata.createIndex({ avgRating: -1 })
< avgRating_-1

```

Step 15 — Check Rating Index

```

> db.book_metadata.find({
  $text: { $search: "Digital" }
}).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'BookFlow.book_metadata',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [
          'digit'
        ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
      }
    }
  }
}

```

```

      dupsTested: 0,
      dupsDropped: 0,
      peakTrackedMemBytes: 0
    }
  },
  queryShapeHash: '79E1597FBF1E7E0551896CC4A5A0604E417E5E37F937A039ABC2D303F8513AD',
  command: {
    find: 'book_metadata',
    filter: {
      avgRating: {
        '$gt': 4
      }
    },
    '$db': 'BookFlow'
  },
  serverInfo: {
    host: 'DESKTOP-H01S07A',
    port: 27017,
    version: '8.3.8',
    gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
  },
  serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
  },
  ok: 1
}

```

Step 16 — Display All Indexes

```

> db.book_metadata.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_ ' },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },
    name: 'reviews.comment_text',
    weights: { 'reviews.comment': 1 },
    default_language: 'english',
    language_override: 'language',
    textIndexVersion: 3
  },
  { v: 2, key: { formats: 1 }, name: 'formats_1' },
  { v: 2, key: { avgRating: -1 }, name: 'avgRating_-1' }
]

```

Final Output:

Week-3_Practical > BookFlow > book_metadata

Documents 4

Aggregations

Schema

Indexes 4

Validation



Type a query: { field: 'value' } or [Generate query](#) ↗



25

1 - 4

```
_id: ObjectId('6a85b3095182dc0f9d9a1bd8')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
  formats: Array (3)
  specs: Object (5)
  avgRating: 4.6
  reviews: Array (2)
```

```
_id: ObjectId('6a85b3095182dc0f9d9a1bd9')
title: "Ocean of Code"
author: "A. Fernandes"
genre: "Technology"
publishedYear: 2021
  formats: Array (2)
  specs: Object (3)
  avgRating: 4.2
  reviews: Array (2)
```

```
_id: ObjectId('6a85b3095182dc0f9d9a1bda')
title: "Whispers of Yesterday"
author: "S. Kapoor"
genre: "Fiction"
publishedYear: 2019
  formats: Array (2)
  specs: Object (2)
  avgRating: 3.8
  reviews: Array (2)
```



```
_id: ObjectId('6a85b3095182dc0f9d9a1bdb')
title: "Quantum Horizons"
author: "R. Menon"
genre: "Science"
publishedYear: 2023
  formats: Array (2)
  specs: Object (4)
  avgRating: 4.9
  reviews: Array (2)
```